

Consider the experience of legendary trader Jesse Livermore in the early 20th century. Despite his extraordinary success, Livermore noted in his memoirs that his biggest losses came when he deviated from his trading rules due to emotional decisions—a problem that algorithms don't face.

?? Did You Know? During the 1929 stock market crash, many orders couldn't be executed because telephone lines were jammed with panicked traders trying to sell their stocks. Modern algorithmic trading systems can execute millions of orders per second, making such bottlenecks virtually impossible today.

The Rise of Algorithmic Trading

The shift toward algorithmic trading began gradually in the 1970s and accelerated dramatically with the technological boom of the 1990s and 2000s. This transition was driven by several key developments:

1.
Digitization of exchanges: The NYSE and NASDAQ moved from physical trading floors to electronic trading platforms.
2.
Increased computing power: Moore's Law enabled more complex calculations to be performed rapidly.
3.
Financial theory advancements: Mathematical models like the Black-Scholes options pricing model created frameworks for systematic trading.
4.
Regulatory changes: Regulations like the SEC's Regulation NMS in 2005 mandated that U.S. stock orders be routed to the venue offering the best price, creating opportunities for algorithmic arbitrage.

By the early 2000s, firms like Renaissance Technologies were using complex mathematical models to identify market inefficiencies and execute trades automatically.